6. A small electric torch is powered by a single cell which supplies 1.6 J of energy per coulomb of charge passing through the cell. When the torch is switched on, the cell supplies a constant current of 0.50 A to bulb X. The potential difference across the bulb is 1.2 V.

(a) Show that the internal resistance, *r*, of the cell is 0.80 Ω.

[2]

(b) The bulb X is replaced by another bulb Y, which draws a current of 0.30 A. Calculate the potential difference across bulb Y.

potential difference = ... V [2]

(c) Calculate the power lost in the cell when connected to bulb Y.

power lost = ... W [2]

(d) When bulb Y is replaced by bulb Z, the power drawn from the cell is maximum. Determine the maximum current that can be drawn from the cell.

maximum current = ... A [2]